

Generalization of the Limit Determinant

For any $k \times k$ complex or real matrix A , we can evaluate expressions of the form:

$$\det \left(\lim_{n \rightarrow \infty} \left(I + \frac{A}{n} \right)^n \right)$$

Theorem

Using the properties of the matrix exponential and Jacobi's formula, we have the fundamental identity:

$$\det \left(\lim_{n \rightarrow \infty} \left(I + \frac{A}{n} \right)^n \right) = e^{\text{Tr}(A)}$$

Rigorous Justification

1. **The Matrix Exponential:** The limit expression inside the determinant is the standard definition of the matrix exponential, denoted as e^A :

$$\lim_{n \rightarrow \infty} \left(I + \frac{A}{n} \right)^n = e^A$$

2. **Continuity of the Determinant:** Because the determinant is a continuous function (it is a polynomial of the matrix entries), we can pass the determinant operation through the limit:

$$\det \left(\lim_{n \rightarrow \infty} \left(I + \frac{A}{n} \right)^n \right) = \det(e^A)$$

3. **Eigenvalue Relationship:** Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be the eigenvalues of A , accounting for algebraic multiplicities. By the spectral mapping theorem, the eigenvalues of the matrix e^A are $e^{\lambda_1}, e^{\lambda_2}, \dots, e^{\lambda_k}$.
4. **Determinant and Trace Invariants:** The determinant of a matrix is the product of its eigenvalues, and the trace (Tr) is the sum of its eigenvalues. Therefore:

$$\det(e^A) = \prod_{i=1}^k e^{\lambda_i} = e^{\sum_{i=1}^k \lambda_i} = e^{\text{Tr}(A)}$$

Application to the Specific Problem

We are given the specific matrix:

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

First, calculate the trace (the sum of the elements on the main diagonal):

$$\text{Tr}(A) = 0 + 0 = 0$$

Finally, apply the generalized theorem to evaluate the determinant of the limit:

$$\det \left(\lim_{n \rightarrow \infty} \left(I + \frac{A}{n} \right)^n \right) = e^{\text{Tr}(A)} = e^0 = 1$$